

App Development Project Report

1. Student Details

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About Me: I am a student focused on developing full-stack solutions for educational environments. I am passionate about creating systems that improve institutional efficiency through automation and intuitive user interfaces.

2. Project Details

Project Title: ShaanU Institutional Placement Portal

Problem Statement: Traditional placement processes are often fragmented, relying on manual coordination between students, companies, and the placement office. This leads to missed deadlines, communication gaps, and difficulty in tracking candidate progress. The ShaanU Portal aims to provide a unified ecosystem to streamline recruitment drives, interview scheduling, and offer management.

Approach: The application utilizes a **Flask-based RESTful API** for the backend and a reactive **Vue.js** frontend. It implements a strict **Role-Based Access Control (RBAC)** system for Admins, Students, and Companies. The architecture is modular, separating identity management (Gatekeeper), institutional oversight (Controller), and individual user workflows.

3. AI/LLM Declaration

I used **Gemini Flash** to assist in the following tasks:

- Drafting documentation for complex API routes.

The extent of AI usage is approximately **3%**, primarily focused on structural boilerplate and documentation. All core business logic, database migrations, and frontend-backend integration were performed manually.

Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework and API routing.
Vue.js	Reactive frontend framework for the user dashboards.
SQLAlchemy	ORM for database management and relationship mapping.
Flask-JWT-Extended	Secure token-based authentication and RBAC.
Tailwind CSS	Utility-first styling for a premium institutional UI.
SQLite	Lightweight database for persistent data storage.
Redis / Cache	In-memory caching for administrative metrics.
Chart.js	Visualization of placement trends and analytics.

4. Database Schema / ER Diagram

Tables:

1. **IdentityRegistry**: Stores core auth credentials (id, email, password_hash, role, is_active).
2. **StudentProfile**: Extended details for candidates (id, registry_id, full_name, cgpa, resume_path).
3. **CompanyProfile**: Corporate partner details (id, registry_id, legal_name, industry, status).
4. **PlacementOpportunity**: Stores drive details (id, employer_id, job_title, package, status).
5. **PlacementWorkflow**: Tracks student applications (id, student_id, opportunity_id, current_stage).
6. **InterviewSession**: Manages scheduling (id, workflow_id, proposed_time, link, status).
7. **Notification**: Real-time alerts for all users (id, receiver_id, message, type).

Relationships:

- **User to Profile:** One-to-One (IdentityRegistry ↔ Student/Company).
- **Company to Drive:** One-to-Many (CompanyProfile ↔ PlacementOpportunity).
- **Drive to Application:** One-to-Many (PlacementOpportunity ↔ PlacementWorkflow).
- **Application to Interview:** One-to-Many (PlacementWorkflow ↔ InterviewSession).

5. API Resource Endpoints

Endpoint	Method	Description
/gatekeeper/verify-access	POST	Authenticate user and return JWT access token.
/controller/dashboard/metrics	GET	Fetch institutional stats for the Admin dashboard.
/enterprise/publish-drive	POST	Create a new recruitment drive for Admin approval.
/candidate/application-tracking	GET	Fetch the lifecycle history of a student's applications.
/enterprise/interviews/schedule	POST	Dispatched interview invitations to candidates.
/candidate/interviews/respond	POST	Accept or request a reschedule for an interview.

6. Architecture and Features

Architecture Overview:

- `app.py`: Main entry point and Flask configuration.
- `/backend/routes/`: Modularized APIs (`admin.py`, `company.py`, `student.py`, `auth.py`).
- `/backend/models/`: SQLAlchemy class definitions.
- `/frontend/components/`: Vue.js single-file components for different roles.
- `/static/`: Stores generated PDF offer letters and uploaded resumes.

Implemented Features:

- **Multi-Role Dashboards:** Custom interfaces for Admins, Companies, and Students.
- **Drive Moderation:** Admin workflow to authorize or deny corporate drives.
- **Interview Orchestration:** Two-way confirmation loop for scheduling.
- **Offer Generation:** Automated PDF creation for selected candidates.
- **Ecosystem Search:** Real-time filtering across all institutional records.
- **Institutional Security:** Blacklisting capabilities for suspicious accounts.

7. Video Presentation

Drive Link: https://drive.google.com/file/d/1sa1dHdykCr5FwdN07tP_f5oBrBvFA54h/view?usp=drive_link